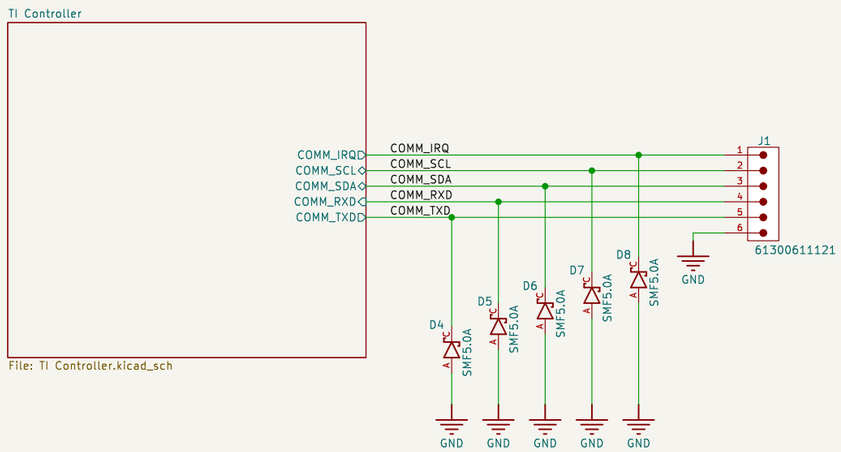
ULTRASONIC FLOW METER CONTROLLER

Revision History		
Revision	Revision Notes	Date
001	Initial Release	14/SEP/2025
002	Client comments updated TVS diodes Added	17/SEP/2025

Changes from EVM430-FR6043	
Page	Changes
1	-Removed HID Brige, USB Hub, ezFET and U_EVM430-FR6043_HW sections -Removed J2, J3, J4 and J12 USB Connector -Changed J1 connectors
2	-Removed Booster Pack, Booster Pack Muxing, MTIF and LCD sections -Removed Y3 Resonator, S2 and S4 buttons -Removed S5 Switch , J31 connector -Mount Y4 crystal -Removed J32 connector and Short with 0E Resistors -Added 3.3V LDO and updated Power nets
3	-Removed JP1, JP2, JP3 Connectors and U5(+5V DCDC REG) section -Removed J9 Connector and Short with 0E resistors -Removed J10 & J11 Connectors and Terminated with Testpoints -Removed J5 & J6 Connectors and Terminated with Testpoints -Removed JP4 and JP5 Connectors and Short with 0E Resistors
4-7	-Pages removed from schematics



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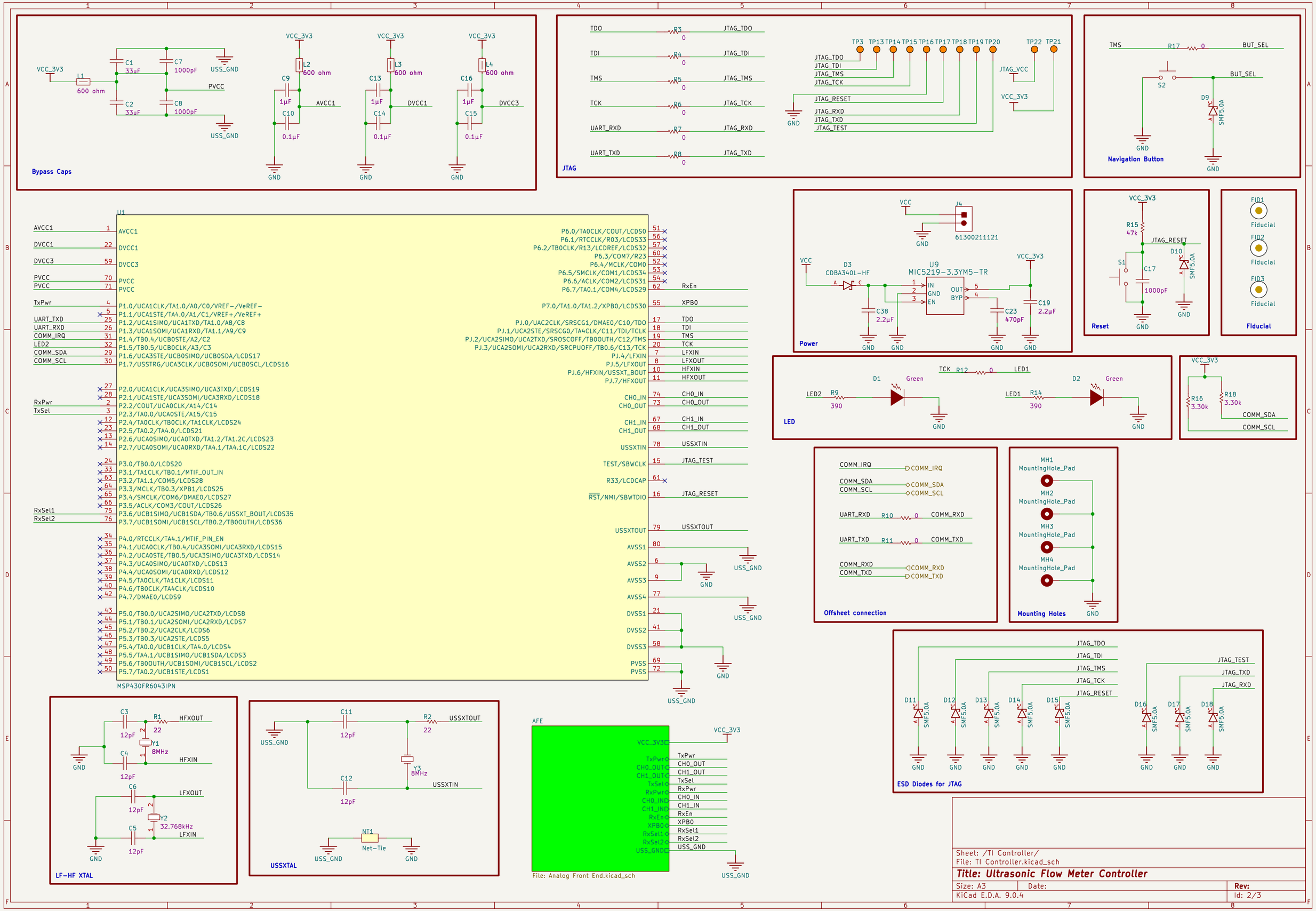
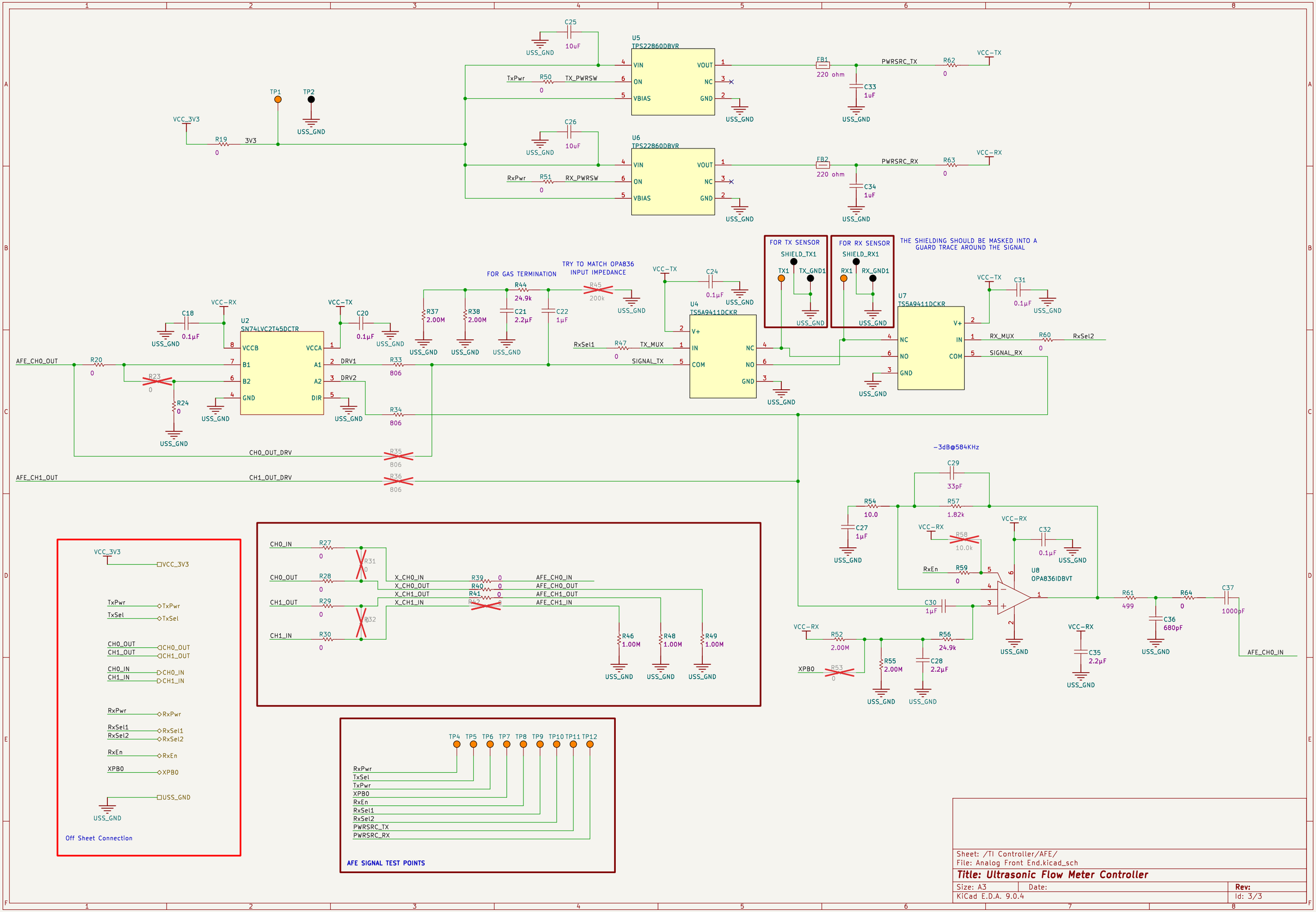
Title: Ultrasonic Flow Meter Controller

Size: A3	Date: 2025-09-13	Rev: 1
KiCad E.D.A. 9.0.4		Id: 1/3


Bypass Caps
VCC_3V3
L1
600 ohm
C1
33uF
C2
33uF
C7
1000pF
C8
1000pF
USS_GND
PVCC
L2
600 ohm
L3
600 ohm
L4
600 ohm
C9
1µF
C10
0.1µF
AVCC1
C13
1µF
C14
0.1µF
DVCC1
C16
1µF
C15
0.1µF
DVCC3
GND
JTAG
TDO
TDI
TMS
TCK
UART_RXD
UART_TXD
R3
R4
R5
R6
R7
R8
0
JTAG_TDO
JTAG_TDI
JTAG_TMS
JTAG_TCK
JTAG_RXD
JTAG_TXD
TP3
TP13
TP14
TP15
TP16
TP17
TP18
TP19
TP20
TP22
TP21
JTAG_RESET
JTAG_TEST
JTAG_VCC
VCC_3V3
Navigation Button
TMS
R17
BUT_SEL
S2
D9
SMF5.0A
U1
MSP430FR6043IPN
AVCC1
DVCC1
DVCC3
PVCC
TxPwr
UART_TXD
UART_RXD
COMM_IRQ
LED2
COMM_SDA
COMM_SCL
RxPwr
TxSel
RxSel1
RxSel2
RxEn
XPB0
TDO
TDI
TMS
TCK
LFXIN
LFXOUT
HFXIN
HFXOUT
CH0_IN
CH0_OUT
CH1_IN
CH1_OUT
USSXTIN
USSXTOUT
JTAG_TEST
JTAG_RESET
AVSS1
AVSS2
AVSS3
AVSS4
DVSS1
DVSS2
DVSS3
PVSS
Power
VCC
J4
61300211121
D3
CDBA340L-HF
U9
MIC5219-3.3YM5-TR
IN
GND
EN
OUT
BYP
C38
2.2µF
C23
470pF
C19
2.2µF
Reset
R15
47k
S1
C17
1000pF
D10
SMF5.0A
Fiducial
FID1
FID2
FID3
LED
D1
Green
D2
Green
LED2
R9
390
TCK
R12
LED1
R14
R16
3.30k
R18
3.30k
COMM_SDA
COMM_SCL
Offsheet connection
COMM_IRQ
UART_RXD
R10
COMM_RXD
UART_TXD
R11
Mounting Holes
MH1
MH2
MH3
MH4
MountingHole_Pad
ESD Diodes for JTAG
D11
D12
D13
D14
D15
D16
D17
D18
LF-HF XTAL
C3
12pF
R1
22
HFXOUT
Y1
8MHz
C4
HFXIN
C6
LFXOUT
Y2
32.768kHz
C5
LFXIN
USSXTAL
C11
R2
USSXTOUT
Y3
8MHz
C12
USSXTIN
NT1
Net-Tie
AFE
File: Analog Front End.kicad_sch
Sheet: /TI Controller/
File: TI Controller.kicad_sch
Title: Ultrasonic Flow Meter Controller
Size: A3
Date:
Rev:
KiCad E.D.A. 9.0.4
Id: 2/3



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Title: Ultrasonic Flow Meter Controller

Size: A3	Date:	Rev:
KiCad E.D.A. 9.0.4		Id: 3/3